

Rakshak S Barkur

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EDUCATION

BMS Institute of Technology & Management, Bengaluru

BE in Department of Artificial Intelligence and Machine Learning

Relevant Coursework: DSA, AI, DBMS, OS, OOPS with Java, CN, MongoDB & Git.

Sep 2023 - May 2027

CGPA - 8.53

TECHNICAL STRENGTHS

Languages : C/C++, Python, Kotlin, HTML, CSS, JavaScript & Java

Tools : VSCode, LaTeX, Excel, Jupyter Notebook & MATLAB

Databases : SQL, MongoDB, Qdrant, FAISS, PostgreSQL & Redis

Libraries : STL (C++), LlamaIndex, Hugging Face Transformers, Streamlit, Flower, Numpy, Scikit-Learn, Pandas, TensorFlow, OpenCV & Matplotlib

Certifications : C++, Python, DSA, Google AI Essential, Google Prompt Engineering, CP, MySQL & SAP Backend

EXPERIENCE

TestAing - AI/ML Intern

Mar'26 - Jul'26

- Architected a multi-agent framework utilizing LangGraph and FastAPI, integrating a supervisor node that routes complex tasks dynamically to four domain-specific specialist agents with parallel execution paths and a contextual confidence-scoring policy engine.
- Engineered real-time, bidirectional state synchronization between a React frontend and backend agents using the AG-UI protocol over SSE, leveraging CopilotKit and RFC 6902 JSON Patch deltas to stream predictive state updates.
- Designed a flexible frontend rendering layer that supports both pixel-perfect Controlled React components via CopilotKit and schema-driven Declarative GenUI templates adhering to Google's A2UI specification.

Conatus Bharat - Full Stack Development, Intern

Sep'25 - Mar'26

- Designed and maintained a digital education platform enabling structured content delivery using TypeScript, NestJS for backend and frontend using Next.js, TypeScript and React, styled with TailwindCSS and used PostgreSQL for the database.
- Architected and refactored platform into a modular, scalable system with RESTful APIs and JWT/bcrypt authentication, reducing average latency by 88% (from 25ms to 3ms).
- Optimized PostgreSQL database performance by designing indexes, constraints and schema improvements, increasing throughput.

PROJECTS

Early Detection of Neonatal Sepsis using Federated Learning /Neonatal-Sepsis

Accepted into ICICI'26

- Implemented a privacy-preserving federated learning framework for early neonatal sepsis prediction, enabling secure multi-hospital EHR training using Flower and FedAvg, ensuring HIPAA- and GDPR-compliant collaboration.
- Developed GRU-D and Transformer time-series models on 40 NICU clinical features, improving AUROC from 0.829 $\rightarrow$ 0.893 and accuracy from 73.9% $\rightarrow$ 94.7% to enhance early detection reliability.
- Strengthened clinical decision support by increasing precision from 0.187 to 0.712 ($\sim$ 280) and reducing the false-positive rate from 0.2422 to 0.0147, minimising alarm fatigue and unnecessary interventions.

RAG: Agentic Compliance RAG Engine /RAG

- Architected a Python Agentic RAG system featuring a multi-stage reasoning loop that autonomously enhanced query effectiveness, improving answer relevance for over 1,000 unique data sources.
- Engineered a high-precision retrieval engine using Qdrant, implementing Hybrid Search (Dense + BM25) fused via Reciprocal Rank Fusion (RRF) and FlashRank cross-encoder reranking to achieve superior accuracy for both semantic and exact-keyword queries.
- Optimised performance and cost through a tiered LLM strategy, leveraging local Mistral (Ollama) for reasoning and fact-checking while reserving Gemini 2.0 Flash for final synthesis, reducing cloud API consumption by $\sim$ 80%.

Folibuddy: AI-Powered Resume to Portfolio Generator /Folibuddy

Live Demo @folibuddy

- Automated conversion of PDF resumes into deployable personal websites, implementing backend in Python, backed with FastAPI and Google Gemini API to orchestrate parsing and site generation.
- Implemented structured LLM outputs for resume parsing and designed an editable input interface enabling users to review & personalise extracted content before publishing the site to a custom domain/URL.

EXTRA CURRICULAR ACTIVITIES

Competitive Programming:

- `</>` LeetCode : Solved 450+ problems, Peak Rating 1953.
- `</>` GeeksforGeeks : Solved 700+ problems

Positions of Responsibility: Serving as Chair & VP of the Alterino, a technical Club at BMSIT, leading event organisation and logistics.

Open Source Contributor (GirlScript Summer of Code): Actively contributing to open-source projects and currently ranked among the Top 200 contributors globally.